Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH (AUTOMOBILE ENGINEERING) END SEMESTER EXAMINATIONS, NOV 2018

Ergonomics in Automotive Design [AAE 4077]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1)		Describe how ergonomic research studies contribute to automotive design. Compare descriptive, experimental, and evaluative research methods in this context.	(5)
	A)		
	B)	List three advantages of touchscreen controls in vehicles. Also, mention one drawback associated with this type of control.	(3)
	C)	Briefly explain what is cumulative trauma disorder (CTD).	(2)
2)		Discuss the key biomechanical considerations in automotive seat design for reducing stress on the L5/S1 region of the spine.	(5)
	A)		
	B)	Why are both plane and convex mirrors often used on trucks? What specific advantage does this combination provide?	(3)
	C)	What is the significance of the seating reference point (SgRP)?	(2)
3)		Analyze how occupant packaging impacts driver comfort, safety, and accessibility within the vehicle.	(5)
	A)		
	B)	Compare the differences between monocular, ambinocular, and binocular fields of view. Why is binocular vision important for drivers?	(3)
	C)	Describe why "flatter" seatbacks are preferred for long-term comfort.	(2)
4)		Describe the importance of the "Location Expectancy Principle" and the "Frequency of Usage Principle" when designing the layout of controls in a vehicle. How do these	(5)

principles contribute to safety and usability?

A)

B) Describe three specific issues that drivers with short legs may encounter when entering or exiting a vehicle. Discuss how these issues impact comfort and accessibility. (3)

C) What is the role of seat bolsters in entry and exit comfort? Provide one benefit and one potential drawback. (2)

5) Discuss the advantages and disadvantages of using head-up displays (HUDs) in vehicles. Mention at least two of each and provide a real-world example to support your answer. (5)

A)

B) Explain the advantages and disadvantages of using voice controls in vehicles. (3)

C) Describe the purpose of creating a 35-degree down-angle cone when designing control and display locations. Why is this placement advantageous? (2)

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